

We use the mechanics of materials to build **regenerative scaffolds for bone and heart repair** and to design **multifunctional composites for defense and electronics**, combining nanomechanical experiments, manufacturing, and multiscale simulation.

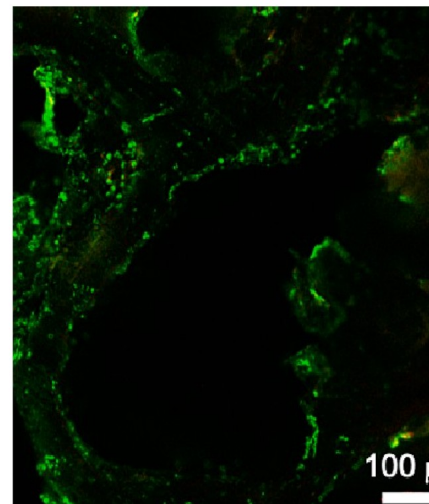
PRIME FOCUS 1

Regenerative engineering

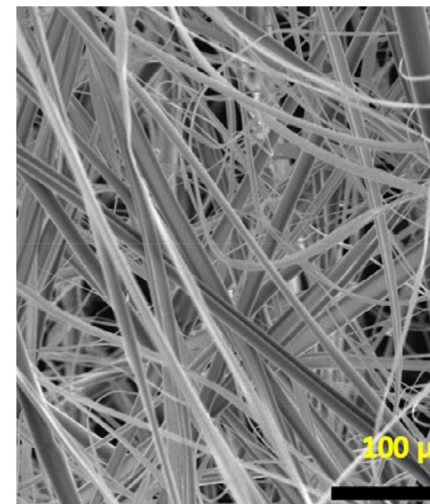
Scaffolds and devices for bone and heart repair



Open-source bioreactor for loading bone tissue



Live/dead staining of cultured bone

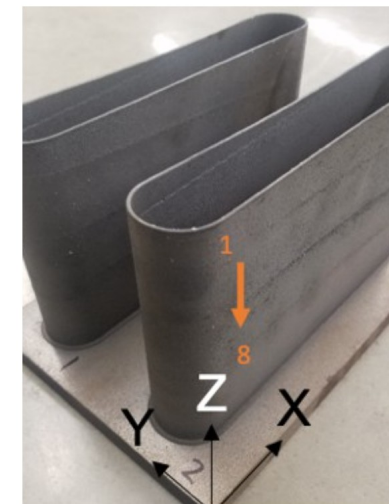


Electrospun cellulose fiber scaffold

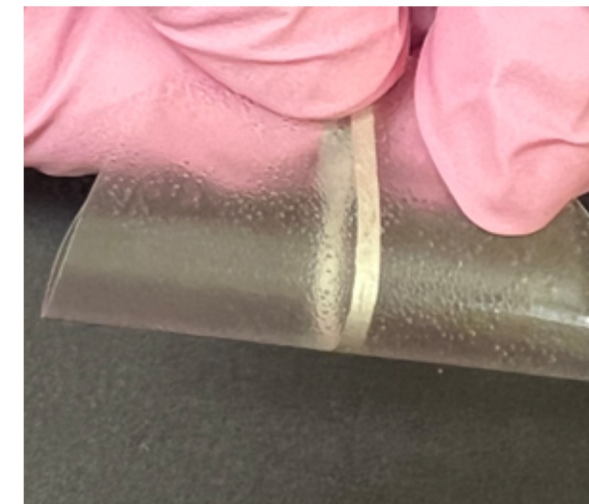
PRIME FOCUS 2

Multifunctional composites

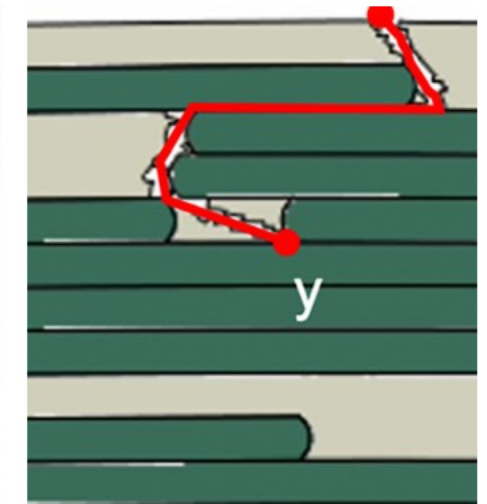
New materials for defense and electronics



3D-printed GRCo-42 alloy for rocket engines



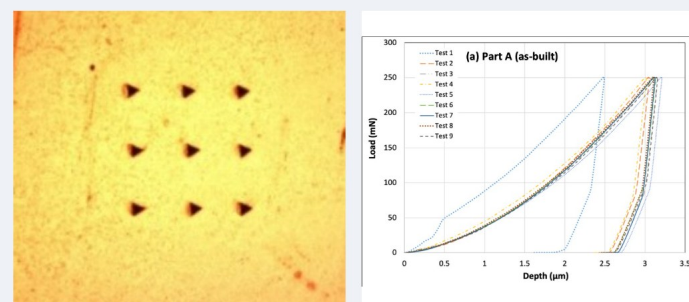
Flexible electronic substrate



Crack path in a MXene-polymer nanocomposite

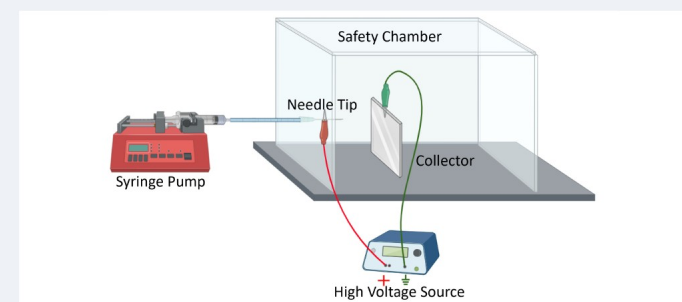
One mechanics toolkit

Nanomechanical experiments



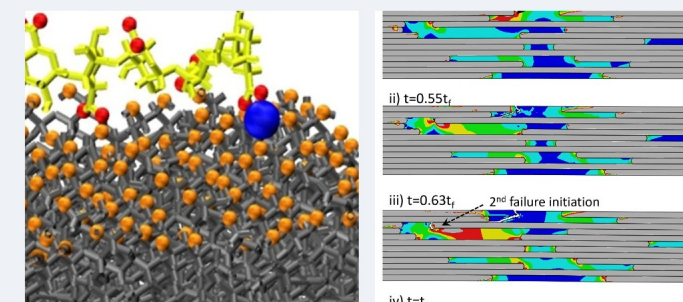
Indent arrays and load-depth curves

Manufacturing



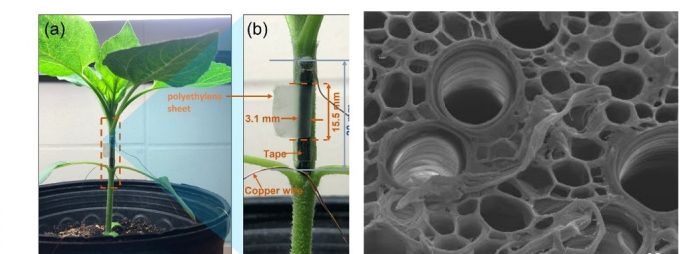
Lab-built electrospinning for fiber scaffolds

Multiscale simulation



Molecular dynamics to finite element models

ALSO APPLIED TO Plant health and precision agriculture



Stem growth sensor; xylem wall mechanics